

Serving Cut Leaves in the $\forall\text{PO}$ -Free Fragment of $\text{ALCl}_{\text{RCC5}}$

Support-Generated Labels, Extremal Carriers, and Demand-Carrying Kernels

Formal technical note prepared from the supplied Lean excerpt and probe packet

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Abstract

The remaining vertical obligation in the mixed, $\forall\text{PO}$ -free fragment of $\text{ALCl}_{\text{RCC5}}$ concerns a blocking cut leaf whose complete model type repeats an ancestor's type but whose own $\exists\text{PP}.D$ or $\exists\text{PPI}.D$ witnesses were not expanded. Two natural repairs have already failed: recursively expanding each cut target yields model-prefix-dependent rounds, while borrowing a blocker's witness through a declared edge collapses to self-witnessing or, after excluding self-witnesses, performs no work.

This note develops a different candidate route. First, certificate labels need not equal complete model types: the literal `MultiTierOk` obligations permit support-generated Hintikka labels containing only formulas needed to establish the root and consequences forced by already-labelled universals. Second, a vertical demand admits an order-theoretic dichotomy. Either its argument has an extremal carrier, which serves the demand and cannot repeat the same defect, or there is an infinite chain of carriers of that same argument, from which the existing recurrence machinery can construct a kernel with demand list $[D]$. Consecutive extremal steps in one direction strictly decrease a finite, formula-derived demand spectrum.

A fixed-formula finite-set fence shows why the distinction matters: the complete-type external-witness expansion is forced through $2L + 1$ model nodes as the fence length L grows, whereas a two-external, zero-kernel support certificate suffices for every L . A new negative-control probe checks all non-vacuous `MultiTierOk` fields for this thin certificate. Separate probes find an extremal carrier for all 305 phase-uncovered residual demands in the supplied tower suite, with zero carriers repeating the same triggering existential. These results refute the inference that the measured target-round growth is inherent to every extraction. They do *not* yet prove a global C_0 -computable bound: support generated by universals and mixed `PP/PPI` alternation remains the decisive open proof obligation.

Status and honesty boundary. This is a candidate construction and research note, not a proof of general decidability. The supplied file `lean/EXCERPTS.lean` contains the stated proof bodies but does not compile standalone because its surrounding 31,783-line development was not supplied. Accordingly:

- **[C]** means reported Lean-certified in the supplied project and inspected in the excerpt, but not independently recompiled here;
- **[M]** means proved mathematically in this note from the stated `RCC5` order assumptions;
- **[E]** means reproduced by executing a supplied or newly added probe with its stated controls;
- **[O]** means open, conjectural, or an implementation interface.

The ZIP accompanying this PDF contains the original archive verbatim, all seven probe sources, fresh outputs, the LaTeX source, and a checksum manifest.

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1 Problem boundary

Let I be a strong-equality RCC5 interpretation. The five role atoms are $\{\text{DR}, \text{PO}, \text{EQ}, \text{PP}, \text{PPI}\}$, where EQ is identity, PP is strict proper part, and PPI is its converse. The fragment forbids formulas $\forall \text{PO}.C$ but permits $\exists \text{PO}.C$. The present question concerns only the vertical PP/PPI component of the remaining mixed quadrant.

For a concept C_0 , write

$$\text{mty}_{C_0}^I(x) = \{F \in \text{cl}(C_0) \mid I, x \models F\}$$

for the complete model type used by the supplied extraction. A certificate $T : \text{MultiTier} \beta \kappa$ has external labels $\tau_E : \beta \rightarrow \text{List}(\text{Concept})$, periodic kernel phase labels, and a quotient RCC5 network. The structure `MultiTierOk` records propositional closure, universal propagation, existential fulfilment, and the quotient-frame conditions.

The blocking closure `cutNodes` records complete model types along the current path. If a selected witness repeats a type already in the path, the witness is kept but not recursively expanded. Such a witness is a *cut leaf*. It is therefore present as an external node, yet any existential formulas inherited from its complete type still require services.

Open residue. Given a cut leaf v with $\exists \text{PP}.D \in \text{mty}_{C_0}^I(v)$, place a legitimate service in the certificate without (i) identifying v with another node, (ii) overriding the read-off RCC5 frame with an inconsistent declared edge, or (iii) initiating a number of complete-type expansion rounds controlled by an arbitrary model prefix rather than by C_0 .

2 Certified baseline and failed routes

2.1 Reported certified facts

Lean name	Consequence used in this note
<code>odOfModel</code> , <code>odNet_frame</code>	The model's own PP and DR relations form an order/disjointness structure, and its induced RCC5 network is composition-closed. [C]
<code>cutNodes_stable_typeEnum</code>	The current blocking closure stabilises at fuel $ \text{typeEnum}(C_0) $ for every witness selector. This proves termination of one closure pass, not service of cut leaves. [C]
<code>cutNodes_up_mem</code> , <code>cutNodes_dn_mem</code>	Every vertical witness selected for an expanded node is present in the closure, including a witness retained as a cut. [C]
<code>readoff_ee_all_pp</code> , <code>readoff_ee_all_ppi</code> , <code>readoff_ee_all_dr</code>	Under read-off relations, external universal propagation follows from model truth and the frame. [C]
<code>readoff_e_ex_pp</code> , <code>readoff_e_ex_ppi</code>	Vertical external existential fulfilment reduces to membership of a genuine model witness in the selected set or in a serving kernel phase. [C]

Lean name	Consequence used in this note
<code>chain_or_kernel</code> , <code>rr_covers</code>	An arbitrary ascending demand-following chain either terminates or yields a kernel; persistent demand lists can be covered round-robin. The supplied <code>chain_or_kernel</code> builds <code>KernelData</code> with an empty demand list. [C]

These facts sharply isolate the issue. With read-off relations, adding a real model point preserves the RCC5 frame and makes universal propagation truthful. The unresolved question is which real points, kernels, or regular occurrences must be represented finitely.

2.2 Why the two rejected answers remain rejected

Complete-type target expansion. Probe `wp116_target_rounds.py` adds and expands targets for unresolved cut leaves. With period fixed at three, the reproduced maximum round count grows with the aperiodic prefix:

Prefix L	4	8	12	18	26	40
Maximum rounds	3	5	6	9	12	≥ 15

At $L = 40$, 19 runs hit the ceiling, so 15 is only a lower bound. The control requiring an unresolved round-zero cut leaf was positive in every shape. [E] This refutes a bound for that extraction strategy based solely on repeated complete types.

Declared-edge borrowing. The historical declared-edge treatment achieved apparent 100% coverage only by allowing a cut leaf to be its own witness. Once $w \neq v$ is enforced, the 305 phase-uncovered cases have no already-selected edge witness; all 305 require a new model point. In the current `wp118_multitier_acceptance.py`, declared edges ON and OFF produce exactly the same failures:

$$e_ex_PP = 37, \quad k_ex_PP = 42, \quad k_ex_PPI = 336.$$

Thus the guarded treatment performs no work in the kernel-bearing class. [E]

Warning 2.1. The reproduced `wp117` banner still states that its redesign has an answer, but the same output reports “witness is a NEW node: 305.” The banner is a retired conclusion, not evidence. The accompanying package preserves this verbatim so the inconsistency remains auditable.

3 The architectural slack: labels need not be complete types

Observation 3.1 (No full-type axiom in `MultiTierOk`). The literal definition of `MultiTierOk` contains no field asserting

$$\tau_E(e) = \text{mtv}_{C_0}^I(e) \quad \text{or} \quad \text{phase}(k, a) = \text{mtv}_{C_0}^I(x_{k,a}).$$

Its obligations are conditional on formulas that actually occur in the certificate labels.

Proof. Inspection of the supplied structure shows only clash-freedom, absence of bottom, Boolean decomposition, universal propagation, and existential fulfilment for labelled formulas, together with the quotient-frame condition. There is neither a maximality clause nor a requirement to list every true subformula of C_0 . $\square$

This observation does not itself produce a certificate. It identifies an unnecessary extraction invariant: the current construction equates a certificate label with the complete set of all true closure formulas, including true formulas beneath disjuncts that were not chosen to support C_0 .

3.1 Support-generated labels

Construction 3.2 (Model-guided support saturation). Maintain a finite or regular collection of external occurrences and kernel phase occurrences. Associate each occurrence u with a support label $\Gamma(u) \subseteq \text{cl}(C_0)$. Seed the root occurrence u_0 with C_0 , and close under the following rules.

1. If $C \sqcap D \in \Gamma(u)$, add both C and D to $\Gamma(u)$.
2. If $C \sqcup D \in \Gamma(u)$, add one disjunct true at the model point represented by u .
3. If $\forall r.C \in \Gamma(u)$ and the certificate's read-off relation from u to v is r , add C to $\Gamma(v)$. The analogous rules are applied to all external/kernel edge classes required by **MultiTierOk**.
4. If $\exists r.C \in \Gamma(u)$, record one legal service branch and add C to the serving external or phase label.

No other true formula is inserted merely because it belongs to the represented model point's complete type.

Lemma 3.3 (Truth invariant). *Assume every external service is a genuine model witness, every relation is read off from the model, every disjunction chooses a model-true disjunct, and every kernel phase is represented by a model point on its witnessing chain. Then support saturation preserves*

$$F \in \Gamma(u) \implies I, x_u \models F.$$

Proof. Induct over insertions. The root seed is true by hypothesis. Conjunction and disjunction insertions preserve truth by their semantics. For a universal, read-off relation r and truth of $\forall r.C$ at the source imply truth of C at the target. For an existential, the chosen service is required to be a real r -successor satisfying its argument. The kernel case uses the model points that witness the kernel phases. $\square$

Proposition 3.4 (Conditional obligation discharge). *Suppose support saturation terminates in a finite multi-tier object, the quotient network is a read-off **odNet**, and every labelled existential has a recorded legal service branch. Then the resulting labels satisfy the propositional, universal, and existential fields of **MultiTierOk**.*

Proof. The Boolean fields hold by saturation rules 1–2. The universal fields hold by rule 3 for each edge class. The existential fields hold by rule 4 and the recorded service branch. The frame field is supplied independently by the read-off order/disjointness construction. Kernel period positivity and phase well-formedness remain explicit construction side conditions. $\square$

What this changes, and what it does not. A new witness may still be added as a distinct external node. The difference from target-round expansion is that the node receives only the demanded argument and forced support consequences, not its complete model type. The open theorem is therefore termination or regular representability of the *support-generated* service graph, not termination of repeated complete model-type expansion.

4 Extremal carrier or demand-carrying kernel

The second component changes witness selection. It is order-theoretic and does not rely on type repetition.

Definition 4.1 (Ascending carrier set). For a demand $\exists \text{PP}.D$ true at x , define

$$\text{Car}_D^+(x) = \{y \mid x < y, I, y \models D\},$$

where $x < y$ abbreviates $I.\rho(x, y) = \text{PP}$. A member y is *D-maximal* if no $z \in \text{Car}_D^+(x)$ satisfies $y < z$.

Theorem 4.2 (Extremal carrier or *D*-chain). *Let $<$ be the strict transitive PP order of an RCC5 interpretation. If $I, x \models \exists \text{PP}.D$, then the following case distinction is available:*

1. *there is a D-maximal $y \in \text{Car}_D^+(x)$; or*
2. *there is a sequence $c : \mathbb{N} \rightarrow \Delta^I$ with $c(0) = x$, $c(n) < c(n+1)$ for all n , and $I, c(n) \models D$ for every $n \geq 1$.*

The displayed outcomes can coexist in a branching order: a model may contain both a maximal carrier and a separate infinite carrier branch. The construction uses the first outcome whenever a maximal carrier exists and invokes the chain construction only under the negation of that existence claim.

Proof. The carrier set is nonempty by the existential semantics. If it has a maximal member, choose it. Otherwise every *D*-carrier above x has a strictly higher *D*-carrier. Choose $c(1) \in \text{Car}_D^+(x)$; then use dependent choice to select $c(n+1)$ above $c(n)$ while preserving *D*. Set $c(0) = x$. Strict transitivity keeps every later point above x . $\square$

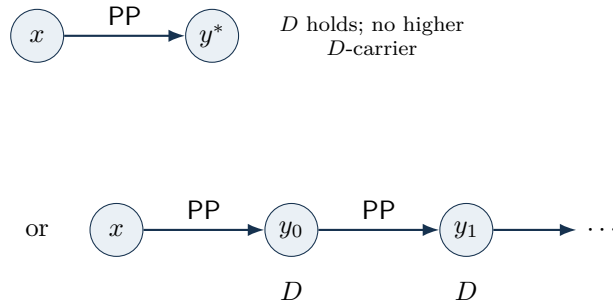


Figure 1: Carrier-specific dichotomy for a demand $\exists \text{PP}.D$.

Lemma 4.3 (An extremal target drops the same defect). *If y is a D -maximal member of $\text{Car}_D^+(x)$, then $I, y \not\models \exists \text{PP}.D$.*

Proof. Otherwise existential semantics provides z with $y < z$ and $I, z \models D$. Transitivity gives $x < z$, so $z \in \text{Car}_D^+(x)$, contradicting maximality. $\square$

Proposition 4.4 (Demand-carrying kernel from the non-extremal branch). *Assume the supplied recurrence theorem `kernel_of_chain`. From the second branch of definition 4.2, one can construct*

$$\text{KernelData } I \ C_0 \ [D] \ L_0 \ \text{true } x.$$

In particular, the kernel's `ccovers` field is discharged without a persistence guard.

Proof. Apply the existing chain recurrence theorem to c and choose its returned indices $i > 0$ and $p > 0$. Copy the domain, step, root, base, positivity, and repeated-type fields into `KernelData`. For the sole demand $D \in [D]$, choose phase offset $b = 0$. Since $i > 0$, the construction of c gives $I, c(i) \models D$, hence $D \in \text{nty}_{C_0}^I(c(i + 0))$, and $0 < p$. Thus `ccovers` holds. No round-robin persistence assumption is needed because every positive chain point carries D . $\square$

The descending mirror uses the converse strict order and produces a descending D -carrying kernel for $\exists\text{PPI}.D$. The excerpt explicitly notes that the current `chain_or_kernel` theorem is ascending only, so its Lean mirror is a separate formalization task. [O]

4.1 A finite one-direction rank

Let

$$\text{Arg}_{\text{PP}}(C_0) = \{D \mid \exists \text{PP}. D \in \text{cl}(C_0)\}$$

and define the ascending demand spectrum

$$U(x) = \{D \in \text{Arg}_{\text{PP}}(C_0) \mid I, x \models \exists \text{PP}. D\}.$$

Lemma 4.5 (Monotonicity and strict extremal descent). *If $x < y$, then $U(y) \subseteq U(x)$. If, in addition, y is selected as a D -maximal carrier for a demand $D \in U(x)$, then $U(y) \subsetneq U(x)$.*

Proof. For $E \in U(y)$, choose z with $y < z$ and $I, z \models E$. Transitivity yields $x < z$, so $E \in U(x)$. For strictness, definition 4.3 gives $D \notin U(y)$ while $D \in U(x)$. $\square$

Corollary 4.6 (Bound on consecutive extremal ascending services). *Any sequence in which every step is an extremal PP service has length at most $|\text{Arg}_{\text{PP}}(C_0)|$. The descending mirror is bounded by $|\text{Arg}_{\text{PPI}}(C_0)|$.*

Warning 4.7. This is not yet a global rank. A PP step may enlarge the downward demand spectrum, and a later PPI step may enlarge the upward spectrum. Mixed alternation can therefore evade a simple lexicographic combination. Support labels remove demands that arose only from complete-type overlabelling, but a proof is still needed for alternation generated by formulas genuinely in the support.

5 A fixed-formula fence separating complete types from support

This family gives a concrete model-normalisation witness: the same C_0 admits arbitrarily long satisfying models on which complete-type expansion grows, yet also admits a constant-size certificate extracted from each model.

5.1 Finite-set RCC5 model

Fix a common element c . For $0 \leq i \leq L$, let

$$X_i = \{c, x_i\},$$

and for $0 \leq i < L$, let

$$Y_i = \{c, x_i, x_{i+1}, y_i\}.$$

RCC5 relations are derived from set semantics. Thus $X_i \text{PP} Y_i$ and $X_{i+1} \text{PP} Y_i$; all other distinct pairs not forced by these inclusions overlap properly.

Use atoms R, A_0, A_1, B_0, B_1 . Make R true only at X_0 , alternate A_0, A_1 over the X_i , and alternate B_0, B_1 over the Y_i . Define

$$\begin{aligned} F_0 &= \exists \text{PP}. B_0, & F_1 &= \exists \text{PP}. B_1, \\ G_0 &= \exists \text{PPI}. A_1, & G_1 &= \exists \text{PPI}. A_0, \\ J &= F_1 \sqcap (G_0 \sqcap G_1), \\ C_0 &= F_0 \sqcap (R \sqcup J). \end{aligned}$$

This fixed concept is $\forall \text{PO}$ -free. It is a vertical stress test rather than a complete mixed-quadrant example, because it contains no $\exists \text{PO}$ demand.

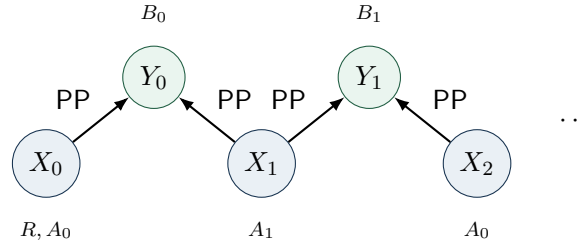


Figure 2: The alternating finite-set fence used by probes wp120 and wp121 . Arrow direction denotes proper-part inclusion.

Proposition 5.1 (Complete-type external-witness expansion traverses the fence). *Any read-off closure over the displayed finite model that labels each selected external with its complete C_0 -model type and fulfils every labelled vertical existential by a genuine external model witness must select all $2L + 1$ fence points.*

Proof. At X_0 , F_0 is true and its only B_0 -carrying proper superregion is Y_0 , so Y_0 is selected. At Y_0 , both G_0 and G_1 are true because its proper subregions X_0 and X_1 carry the two alternating A atoms; hence X_1 is selected. Inductively, once X_i is selected, the appropriate F_0 or F_1 in its complete type forces the previously unselected Y_i ; then the two G formulas at Y_i force X_{i+1} . Thus every X_i and Y_i is selected. $\square$

Proposition 5.2 (Constant-size support certificate). *For every $L \geq 1$, the same model contains a two-external, zero-kernel certificate with labels*

$$\begin{aligned} \Gamma(X_0) &= \{C_0, F_0, R \sqcup J, R\}, \\ \Gamma(Y_0) &= \{B_0\}, \end{aligned}$$

and read-off relation $E(X_0, Y_0) = \text{PP}$.

Proof. The root conjunction is decomposed into F_0 and $R \sqcup J$; the true left disjunct R is selected. The sole labelled existential F_0 is served by the distinct point Y_0 , whose label contains B_0 . There are no labelled universals and no other labelled existentials. The two-point read-off network is an induced subnetwork of the finite-set RCC5 model, so identity, converse, and composition constraints hold. All kernel obligations are vacuous. $\square$

5.2 Measured separation and full-obligation control

L	Model nodes	Complete-type rounds	Complete-type nodes	Support nodes
4	9	2	9	2
8	17	4	17	2
12	25	5	25	2
18	37	8	37	2
26	53	11	53	2
40	81	16	81	2

Probe `wp121_support_multitier_acceptance.py` then checks every non-vacuous field of `MultiTierOk` for the two-node certificate: quotient-frame identity/converse/composition, `e_clash`, `e_nobot`, `e_and`, `e_or`, `ee_all`, and `e_ex`. For each $L \in \{1, 2, 4, 8, 12, 18, 26, 40\}$, removing B_0 from Y_0 yields exactly one `e_ex` failure, while restoring it yields no failures. [E]

The fence is a formal counterexample to the inference “unbounded complete-type target rounds for a fixed formula imply that every certificate extraction for that formula is unbounded.” It is not a proof that support generation is always finite; it isolates complete-type *overlabelling* as one genuine source of spurious model-length growth.

6 Probe evidence and its exact force

All seven probes were executed from a clean extracted copy of the supplied archive. The fresh logs are included in `outputs/reproduced/`.

Probe	Reproduced observation	Permitted conclusion
wp112	Tail continuation 94.0–99.4%; prefix 33.2–55.6% by shape; side 0%. Control failures: 0.	An arbitrary blocked lap often, but not always, continues. Side leaves cannot be repaired by “continue the same chain.”
wp115	Finite read-off certificates pass with declared edges ON and have 37 vertical failures with edges OFF.	The finite model class exercises read-off obligations but cannot validate kernel interactions. Its edge result was superseded.
wp116	Maximum complete-type target rounds grow from 3 to at least 15 as L grows from 4 to 40.	That extraction’s round bound is model-prefix-dependent. No conclusion about all possible certificates follows.

Probe	Reproduced observation	Permitted conclusion
wp117	Phase-or-edge coverage prints 100%, but all 305 edge-only witnesses are new nodes.	Phase coverage is useful; the final “no new node” banner is invalid and is retired.
wp118	Declared edges ON and OFF have identical e_ex/k_ex failures in a kernel-bearing class.	After excluding self-witnesses, the declared-edge treatment does no work.
wp119	All 305 phase-uncovered demands have an extremal carrier; 0 targets repeat the same triggering existential. Random tower sweeps select at most 5 externals and depth at most 3, with controls holding.	Extremal selection defeats the measured prefix effect in this tower class. It is not a MultiTierOk acceptance test and does not settle mixed alternation.
wp120/wp121	For the fixed fence, complete types select $2L + 1$ points while support uses 2; negative control fails and the treatment passes all applicable obligations.	Thin labels are architecturally legitimate for this family. General support termination remains open.

6.1 The 305-demand residue

The exact phase-uncovered residue revisited by wp119 is:

Residual property	Count
Residual demands	305
Has an extremal D -carrier	305
Extremal carrier already selected	0
Extremal carrier is a new real node	305
Target repeats $\exists r.D$	0
No extremal carrier	0

This does not mean that the new target has no other obligations under a complete model type. It means that the target is terminal for the *same signed demand*. Support labels are what prevent unrelated complete-type obligations from restarting the old round mechanism.

6.2 Fixed-formula extremal control

With $C_0 = \exists \text{PP}.A$ fixed while only the irrelevant tower prefix grows, wp119 reports:

L	Nearest-witness rounds	Extremal externals	Depth	Kernels
4	2	2	1	0
8	4	2	1	0
12	6	2	1	0
18	9	2	1	0
26	13	2	1	0
40	20	2	1	0

This is an explicit model-normalisation pattern: select a far extremal carrier rather than accepting the model's nearest-witness prefix.

7 Proposed extraction interface

The current selector always returns a model node, after which `cutNodes` recurses over every vertical existential in its depth-bounded complete type. The proposed selector returns a *service* and the closure operates on support labels.

Listing 1: Lean-style interface sketch; not compiled.

```

inductive VerticalService (I : Interp alpha) (C0 : Concept)
| external
  (y : alpha)                -- genuine, distinct model point
  (hy : I.dom y)
  (edge : Atom)              -- read off from I.rho
  (seed : List Concept)      -- normally [D], then saturated
| kernel
  (D : Concept)
  (K : KernelData I C0 [D] L0 direction root)

structure SupportOccurrence where
  point      : ModelPointOrPhase
  incoming   : Option Atom
  parent     : Option SupportProfile
  label      : Finset Concept

```

7.1 Worklist construction

1. Start with one root occurrence labelled $\{C_0\}$.
2. Saturate Boolean rules and all universal consequences along the current read-off quotient network.
3. For each unserved vertical $\exists r.D$ in a support label:
 - (a) reuse an already selected genuine target when it serves the demand;
 - (b) otherwise choose an r -extremal D -carrier and add it with seed $\{D\}$;
 - (c) if no extremal carrier exists, insert an r -directed kernel whose demand list is $[D]$.
4. Re-saturate because a new read-off node can create new targets for already-labelled universals.

5. Continue until no obligation is new, or replace a repeated support profile by an explicitly justified regular service object.

7.2 Non-negotiable invariants

- **Truth:** every label remains a subset of the represented model point's complete type.
- **Read-off frame:** no declared override changes the model's RCC5 relation; `odOfModel/odNet_frame` remain applicable.
- **No identity collapse:** an external witness is a distinct occurrence and, where required, a distinct model point; no node serves a strict PP/PPI demand with itself.
- **Service provenance:** each existential stores the exact `MultiTierOk` branch that fulfils it.
- **Control before treatment:** every acceptance probe includes a mutation that must fail the targeted obligation.

8 What remains to prove

The proposal reduces the open question but does not eliminate it. The following milestones are ordered from local and likely straightforward to global and high-risk.

1. **Carrier dichotomy in Lean.** Formalise definition 4.2 for PP, using the existing existential witness lemmas and classical choice. $[M \rightarrow C]$
2. **$[D]$ kernel constructor.** Wrap the existing chain recurrence theorem and prove `ccovers` with offset zero, as in definition 4.4. $[M \rightarrow C]$
3. **Descending mirror.** Repeat for PPI or transport through the converse order, matching the project's descending kernel definitions. $[O]$
4. **Support truth and saturation.** Define finite support labels and prove definition 3.3 for every external/kernel edge class. $[M \rightarrow C]$
5. **Support-to-MultiTierOk theorem.** Show that a saturated, service-complete support object induces the literal obligation fields, including `ek_all`, `ke_all`, `kk_pp`, `kk_ppi`, `kq_all`, and `k_ex`. $[O]$
6. **Global regularity theorem.** Prove that the support-generated service graph is bounded by a computable function of C_0 , or that every repeated support context can be compiled into the existing kernel/Q architecture. This is the decisive unresolved step. $[O]$
7. **Mixed-direction stress suite.** Add adversarial formulas in which universals deliberately regenerate alternating PP and PPI existentials. The treatment must be withheld unless a negative control demonstrates that the probe can observe a missing cross-direction service. $[O]$

8.1 Likely finite state, and why it may still be insufficient

A plausible blocking state is not a complete model type alone but a contextual support profile

$$\Pi(u) = (\Gamma(u), \Gamma(\text{parent}(u)), r_{\text{in}}, B_{\text{PP}}(u), B_{\text{PPI}}(u)),$$

where B_{PP} and B_{PPI} record active box consequences. There are computably many such profiles over $\text{cl}(C_0)$. Standard pairwise blocking intuition suggests that a repeated monotone profile should admit regular unfolding. RCC5 composition and total pairwise relations make that intuition insufficient by itself: a proof must show that replacing the repeated suffix preserves every quotient-frame relation to existing externals and kernels.

If the existing linear kernels cannot encode a non-monotone repeated profile, a precise fallback is an explicit *bud/backpointer* service whose semantics is a fresh-copy unfolding, not identity in the finite quotient. This would answer the user’s option (3), but it extends the certificate architecture and therefore requires a new soundness theorem. [O]

9 Decisive positive and negative outcomes

A positive result needs a theorem of the following shape.

Target completeness theorem. There is a computable function B such that, whenever a $\forall\text{PO}$ -free concept C_0 is satisfiable, there exists a finite support-generated **MultiTier** certificate T with a root e_0 , $C_0 \in \tau_E(e_0)$, $\text{MultiTierOk}(T)$, and

$$|\beta| + \sum_{k \in \kappa} p(k) \leq B(C_0).$$

The certificate uses read-off RCC5 relations and serves each vertical demand by an existing target, an extremal thin external, or a demand-carrying kernel.

A route-killing counterexample would instead be a satisfiable fixed C_0 for which every certificate in the unchanged external-plus-linear-kernel architecture requires unboundedly many support-distinct external occurrences or kernels. To be decisive, such a family must quantify over witness choices and thin labels, not merely over nearest-witness or complete-type extraction.

10 Conclusion

The strongest current diagnosis is that the cut-leaf question is partly posed at the wrong abstraction level. A cut leaf owes every demand in its complete model type only because the extraction chose complete model types as certificate labels. The certificate definition itself asks for a supported Hintikka label, not a maximal one.

The most promising repair is therefore the combination:

$$\boxed{\text{support-generated labels}} + \boxed{\text{extremal } D\text{-carrier or } D\text{-carrying kernel}}.$$

The first component removes obligations introduced solely by true but unused closure formulas. The second bypasses arbitrary model prefixes and ensures that an extremal target cannot repeat the same signed demand. A finite, formula-derived rank bounds consecutive extremal moves in either one direction. The fixed fence and the 305-demand residue both support this diagnosis, and the new full-obligation negative-control probe validates the thin certificate on an adversarial family.

The remaining hard question is no longer “how does a complete-type cut leaf serve all demands inherited from its model point?” It is:

Can every support-generated, mixed-direction recurrence be represented by finitely many externals and the current periodic kernels, with a bound computable from C_0 ?

That is a materially narrower and more falsifiable theorem than the original target-round problem.

A Lean proof targets

The package contains `candidates/lean/CANDIDATE_LEMMAS.lean`, a commented interface sketch. The central intended statement is:

```
theorem pp_carrier_extremal_or_covering_kernel
  (hI : RCC5Interp I) (C0 : Concept) (L0 : Nat)
  (x : alpha) (hx : I.dom x) (D : Concept)
  (hD : Concept.ex pp D in mty C0 I x) :
  (exists y,
    I.dom y /\ I.rho x y = pp /\ D in mty C0 I y /\
    forall z, I.dom z -> I.rho y z = pp -> D notin mty C0 I z)
  \/ Nonempty (KernelData I C0 [D] L0 true x)
```

The exact syntax must be adapted to the full development's witness lemmas and list-membership notation. No claim is made that this sketch compiles.

B Reproduction

From the unpacked project root:

```
cd original/probes
python3 wp112_lap_continuation_closed_form.py
python3 wp115_declared_edge_acceptance.py
python3 wp116_target_rounds.py
python3 wp117_kernel_phase_coverage.py
python3 wp118_multitier_acceptance.py

cd ../../candidates/probes
python3 wp119_extremal_carrier_probe.py
python3 wp120_support_label_fence.py
python3 wp121_support_multitier_acceptance.py
```

The convenience script `scripts/run_all_probes.sh` sets the required `PYTHONPATH` so the candidate probes can import the original tower code. The file `MANIFEST.sha256` records checksums for the packaged artifacts.